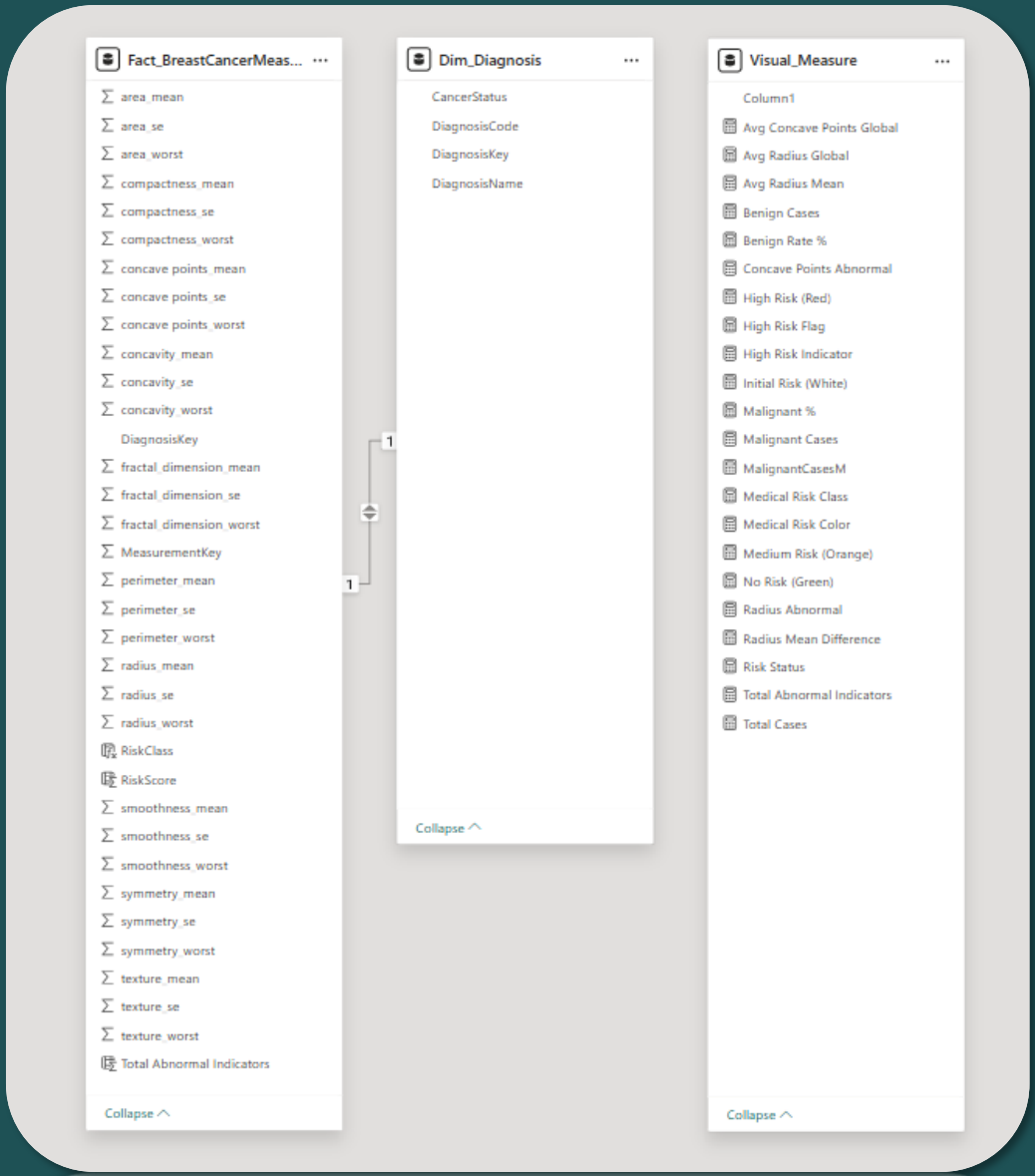
Clinical Risk Stratification for Breast Cancer

Project Goal

- Identify and analyze key diagnostic features related to breast cancer risk
- Stratify patients into No Risk, Initial Risk, Medium Risk, and High Risk categories
- Provide an explainable, data-driven view of how risk levels change across features
- Support clinicians and decision-makers with early risk signals, not just raw data


Fact_BreastCancerMeas...
area_mean
area_se
area_worst
compactness_mean
compactness_se
compactness_worst
concave points_mean
concave points_se
concave points_worst
concavity_mean
concavity_se
concavity_worst
DiagnosisKey
fractal_dimension_mean
fractal_dimension_se
fractal_dimension_worst
MeasurementKey
perimeter_mean
perimeter_se
perimeter_worst
radius_mean
radius_se
radius_worst
RiskClass
RiskScore
smoothness_mean
smoothness_se
smoothness_worst
symmetry_mean
symmetry_se
symmetry_worst
texture_mean
texture_se
texture_worst
Total Abnormal Indicators
Collapse
1
1
Dim_Diagnosis
CancerStatus
DiagnosisCode
DiagnosisKey
DiagnosisName
Collapse
Visual_Measure
Column1
Avg Concave Points Global
Avg Radius Global
Avg Radius Mean
Benign Cases
Benign Rate %
Concave Points Abnormal
High Risk (Red)
High Risk Flag
High Risk Indicator
Initial Risk (White)
Malignant %
Malignant Cases
MalignantCasesM
Medical Risk Class
Medical Risk Color
Medium Risk (Orange)
No Risk (Green)
Radius Abnormal
Radius Mean Difference
Risk Status
Total Abnormal Indicators
Total Cases
Collapse

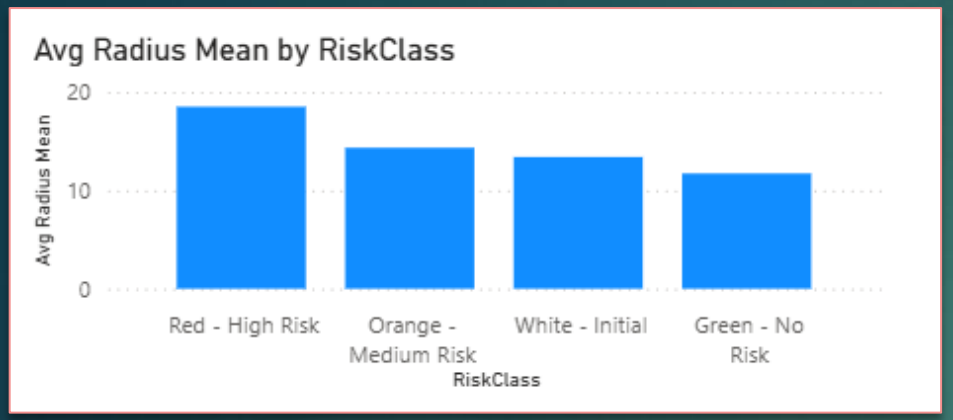
Executive Summary

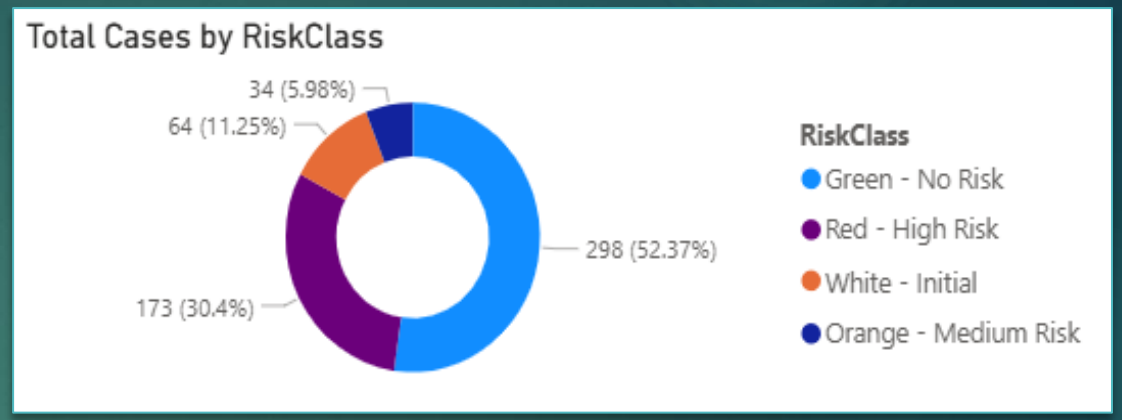
569	357	212
Total Cases	Benign Cases	Malignant Cases

298	64	34	173
No Risk (Green)	Initial Risk (White)	Medium Risk (Ora...	High Risk (Red)

- Out of 569 clinical cases, 212 (37%) are malignant, indicating a significant risk presence
- 207 cases (36%) fall into Medium to High Risk categories, requiring closer clinical attention
- Risk levels show a strong relationship with diagnostic features such as radius, concave points, and texture
- A clear progression is observed from No Risk → Initial → Medium → High Risk, supporting the stratification logic
- The risk framework provides early warning signals, even before final malignancy confirmation

Clinical Risk Stratification





RiskClass	Average of radius_mean	Average of texture_mean	Average of symmetry_mean	Average of concave points_mean
Red - High Risk	18.51	21.28	0.19	0.09
Orange - Medium Risk	14.33	19.90	0.20	0.07
White - Initial	13.39	18.91	0.19	0.05
Green - No Risk	11.72	18.15	0.17	0.02

Critical

Risk Status

This view shows how clinical risk emerges gradually, not suddenly.

When we compare diagnostic features across risk classes, we see a clear progression. As we move from No Risk to Initial, then Medium, and finally High Risk, key indicators such as radius, texture, and concave points consistently increase.
More than one-third of all cases already fall into Medium to High Risk categories. This explains why the overall system flags the situation as Critical.

The key insight here is that risk can be detected early—before a final diagnosis—by observing how multiple features shift together. This stratification framework turns raw measurements into actionable clinical signals.

Storytelling

This project shows that breast cancer risk does not appear suddenly—it evolves.
By analyzing multiple diagnostic features together, we observe a consistent progression from No Risk to Initial, Medium, and High Risk.

A significant share of patients already fall into medium to high risk categories, which is why the overall status is flagged as **Critical**.

While this framework does **not** perform machine-learning prediction, it delivers something equally important: **early, explainable risk signals**. These signals highlight patients who may require closer clinical attention **before** a final diagnosis is made.

In that sense, this work is **ML-ready**—the risk scores and feature patterns established here form a strong foundation for future predictive models.

Most importantly, this framework is designed to **support—not replace—clinical judgment**, transforming raw measurements into clear, actionable insights.